

Desalination plants and renewables combined to solve power and water issues



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ARTICLE INFO

Article history:

Received 10 March 2016

Received in revised form

24 July 2016

Accepted 25 July 2016

Available online 3 August 2016

Keywords:

Desalination

Hydropower

Renewables

Operation model

ABSTRACT

To enhance the security and dispatch ability of a system to supply water and electricity in a cost-effective manner, we propose a model to integrate the operations of a reservoir, hydroelectric power, desalination and wind power. The effect of seasonal energy storage for intermittent wind power is taken into account such that desalination plants can increase power consumption during cold seasons in which wind power is abundant but power demand is small, and can then relieve the burden of water supply from existing reservoirs to enable full operation during peak hours in hot seasons. Our model differs from the combination of an energy-storage system, pumped hydropower, desalination and renewables commonly applied in preceding research. A case study of Taichung city shows that the proposed model can fulfill that water requirement in 2030; the capability of sharing the peak load of existing hydroelectric power units is greatly increased from 398 MW to 1368 MW with an addition of extra 342-MW units, which also eliminate the requirement for 979-MW gas-turbine power fired with natural gas, even though, according to the proposed model, the greater expense from desalination can become compensated by the decreased expense from the power sector.

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1. Introduction

Renewables have played an important role in decreasing carbon emission from the power sector. According to REN21 [1], the renewable fraction of electricity by the end of 2014 was estimated to be 22.8%. Although most renewable electricity still comes from hydroelectric sources (16.6%), the contribution from wind power (3.1%) is no longer negligible and continues to increase. The intermittent nature of wind power, however, becomes a serious issue on integrating much wind power into an existing grid. As wind power increasingly penetrates the power grid, the fluctuation of wind power might cause difficulty in regulation and dispatch of power, and might cause a collapse of the grid system in the worst case [2]. For those countries at latitudes such as that of Germany, the seasonal variation of wind power supply matches the seasonal power demand profile [3], and the installed capacity of wind power can assist to relieve the demand from traditional power units. In

contrast, for those counties at latitudes such as that of Taiwan, the peak demand for power occurs during summer, but the supply of wind power peaks during winter; this seasonal mismatch between supply and demand would result in little relief of traditional power units with increasing wind power.

The climate change affects both the power sector and the water sector. For example, although the annual rainfall apparently remains stable, the intensity of rainfall has increased in recent years; this uneven distribution of rainfall is becoming serious in southern Taiwan [4]. The current system of water supply, especially the reservoirs, will become challenged by the hazards caused by strong rainfall and a water shortage due to a seriously uneven seasonal distribution of rainfall. As most reservoirs in Taiwan face serious problems from silt deposits and as the average remaining effective capacity is only 66%, the water shortage will become worse with the aforementioned problems. From the perspective of the water policy proposed by Taiwan's Water Resource Agency (WRA) [5], 0.5 million cubic meters (Mm^3) per day (182.5 Mm^3 per year) of desalination water will be required in Taiwan by year 2031. Issues of water supply affect also the power supply. As most traditional units for hydroelectric power in Taiwan rely on the water discharged

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Nomenclature

CC	combined cycle
CGT	conventional gas turbine
C_{des}	desalination unit capacity, $\text{Mm}^3 \text{h}^{-1}$ (Million cubic meter per hour)
$C_{hydro, i}$	proposed hydro-turbine capacity of “i” unit, MW
$C_{hydro, design}$	designed flow for proposed Techí hydro-turbine, $\text{Mm}^3 \text{h}^{-1}$
C_{sg}	Shigang reservoir effective capacity, Mm^3
C_{Techi}	Techí reservoir effective capacity, Mm^3
C_{wst}	WST capacity, Mm^3
E_{cgt}	power supported from CGT power plants, MWh
E_{des}	power required by desalination, MWh
E_r	output of hydropower for all units, MWh
$E_{r, i}$	output of hydropower of “i” unit, MWh
E_{wind}	output of wind power, MWh
$EL_{disc, i}$	elevation of hydro-turbine discharge of “i” unit, m
EL_{full}	reservoir full water level, m
EL_{hi}	above this elevation, hydropower is fully operated, m
EL_{hihi}	above this elevation, flushing is required, m
EL_{intake}	elevation of Techí hydropower intake duct, m
$EL_{noflush}$	below this water level, flushing is not possible, m
$EL_{r, i}$	water level of “i” reservoir, m
H_{design}	designed water head for proposed Techí hydro-turbine, m
η_{loss}	proposed hydro-turbine head loss, %
ϕ_{des}	specific energy consumption of desalination, kWh m^{-3}
$\phi_{hydro, i}$	proposed hydro-turbine efficiency at design point for “i” unit, %

$Q_{balance}$	balance of inflow and outflow in Shigang reservoir, $\text{Mm}^3 \text{h}^{-1}$
Q_d	water demand, $\text{Mm}^3 \text{h}^{-1}$
Q_{des}	water supplied by desalination plants, $\text{Mm}^3 \text{h}^{-1}$
$Q_{desilting}$	desilting flow rate, $\text{Mm}^3 \text{h}^{-1}$
Q_{eco}	outflow for ecological consideration, $\text{Mm}^3 \text{h}^{-1}$
$Q_{flushing}$	flushing flow rate, $\text{Mm}^3 \text{h}^{-1}$
Q_{hydro}	water through Techí hydro-turbine, $\text{Mm}^3 \text{h}^{-1}$
$Q_{hydro, design}$	Proposed Techí hydro-turbine flow at design condition, $\text{Mm}^3 \text{h}^{-1}$
$Q_{hydro, max}$	maximal flow for hydropower under a specified water level, $\text{Mm}^3 \text{h}^{-1}$
Q_{Maan}	outflow from Maan unit, $\text{Mm}^3 \text{h}^{-1}$
$Q_{rain, Techi}$	inflow of Techí reservoir, $\text{Mm}^3 \text{h}^{-1}$
Q_{Techi}	outflow from Techí reservoir, $\text{Mm}^3 \text{h}^{-1}$
$Q_{tributary}$	outflow from tributaries downstream of Techí reservoir, $\text{Mm}^3 \text{h}^{-1}$
$Q_{typhoon}$	when inflow is above this flow rate, flushing activity is required, $\text{Mm}^3 \text{h}^{-1}$
$Q_{wst, c}$	water charged to WST, $\text{Mm}^3 \text{h}^{-1}$
$Q_{wst, d}$	water discharged from WST, $\text{Mm}^3 \text{h}^{-1}$
S_{sg}	water amount in Shigang reservoir, Mm^3
S_{Techi}	water amount in Techí reservoir, Mm^3
S_{wst}	water amount in WST, Mm^3
t	time, h
$X_{pattern}$	output ratio of hydropower in specific hour, (0–1, dimensionless)

Subscripts

i 1 through 5 denote units for Techí, Chin-Shan, Guguan, Tianlun and Maan, respectively.

from reservoirs, the capabilities of peak-hour support are limited largely by the water supply duty of the reservoirs.

In recent years, some research has been proposed to improve the integration of renewables and a desalination plant to solve multiple issues. For example, an integration of desalination, a traditional power plant or battery system, wind power or solar power is modeled to enhance the benefits of water supply or the penetration of renewable energies [6–8]. Some research further includes pumped-hydro storage (PHS) or brine-operated PHS in their system [9–11] to improve utilization of intermittent renewables. Gude [12] reviewed various techniques for energy storage to maintain continuous desalination from intermittent renewables; the concept of using desalination as an energy-buffering system to mitigate the intermittence of renewables was investigated [13]. The link between renewable energy and desalination does not inevitably imply a direct use of power generated from renewables for desalination; for example, the annual power required for Sydney's desalination plant [14] is offset by the power production from a wind farm that is purposed to be built for it. Most previous research focused on applications in remote areas or islands and problems related to a short-term intermittence of the renewables. Even for a system that considered the long-term seasonal storage of energy [15], the target was achieved with hydrogen technology rather than with desalination.

Rather than a common combination of a desalination plant, a storage system (either a battery system or PHS) and wind power, we propose an operating model among desalination plants, a reservoir with traditional hydropower and wind power to solve the aforementioned seasonally related issues. The historical data of Techí reservoir and the water demand in Taichung city in Taiwan

serve to demonstrate the feasibility of the model.

2. System description

Fig. 1 shows our studied system: offshore wind turbines,

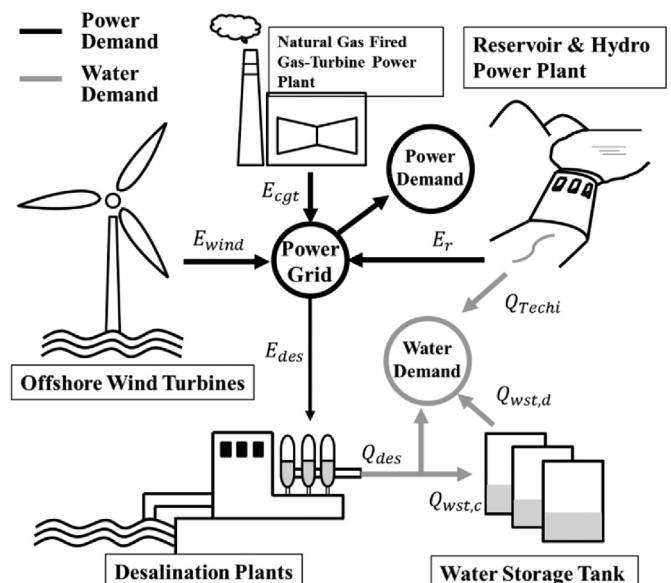
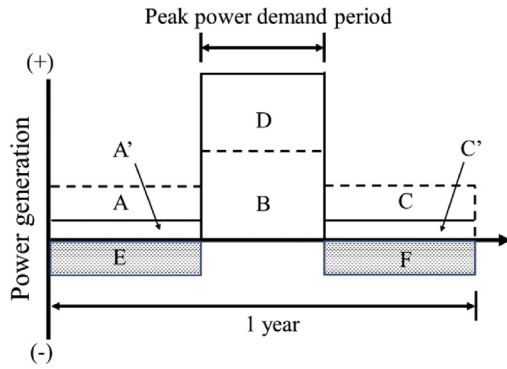


Fig. 1. Power and water supply system in this research.



A+B+C : Hydropower generation pattern of original system

A'+B+C'+D : Hydropower generation pattern of proposed system

E+F : Additional power demand for proposed system

Fig. 2. Patterns of power supply between the proposed model and the original system.

conventional gas-turbine (CGT) power plants, and hydroelectric power plants are included within the system to supply power; desalination plants, water-storage tanks and reservoirs are included within the water-supply system. During hot seasons, hydropower (E_r) from reservoirs becomes fully operated at 11–12 a.m., 1–5 p.m. and 7–8 p.m., which are the typical periods of peak power demand in Taiwan. Beyond these hours, reservoirs maintain an outflow (Q_{Techi}) at a minimum rate to meet the ecological requirement (Q_{eco}). In contrast, the desalination plants operate only at non-peak hours to prevent an increased load at peak hours.

The fruitful wind power (E_{wind}) generated during cold seasons is injected directly into the existing power grid; no power storage system is required. As the proposed system provides additional power demand during the cold season for desalination plants to generate fresh water for the water demand or to charge water-storage tanks (WST), wind power is partly consumed by this demand, whereas the outflow of reservoirs during cold seasons becomes turned to support part of only the water demand.

Fig. 2 shows philosophy of the system operation; sections A, B and C represent the hydropower generation of the original system. With the extra power consumption for desalination plants in sections E and F and less hydropower supply in sections A' and C', the proposed model has an increased capability of hydropower supply during peak-power demand (sections B + D).

3. Data collection and case selection

3.1. Power demand and supply

Most information related to the power generation required in this research can be found from Taiwan Power Company (TPC) [16]. For instance, a typical, hourly based distribution of power generation in Taiwan is shown in Fig. 3. Examination of those peak-load units makes clear that many spare units are required for natural-gas power plants to meet the peak-power demand in summer. In this research, the power-demand pattern in 2011 (Fig. 3) served for all studied cases.

3.2. Offshore wind power

As the first offshore wind turbine (OWT) was still under construction when this research was completed, no datum is available for an OWT in Taiwan. The evaluation of the power production from OWT was simulated on taking data for the wind speed

offshore near the coast of Chiayi county [17], which were converted to the production of power with a curve acquired from wind turbine SWT-3.6-120 (Siemens, 3.6 MW, rotor diameter 120 m) [18]. The simulated power production of a single SWT-3.6-120 wind turbine is shown in Fig. 4; the season of full-load operation is observed generally between October and April. As the development target of capacity and annual production for offshore wind power were set to be 3000 MW and 11,100 GWh, respectively, by BOE in 2025, the wind-speed data were modified in a proportional way (multiplied by 1.14) to meet the announced capacity factor (42.2%).

3.3. Water demand and supply

Because the rate of energy consumption of desalination by reverse-osmosis (RO) is decreasing, this technique is rapidly spreading for the current production of fresh water; the specific power consumption of the state-of-the-art development is less than 2.0 kW h m^{-3} [9]. In this research, we adopted RO desalination, and set the specific energy consumption of desalination (ϕ_{des}) to be 3.3 kW h m^{-3} according to the desalination project proposed in recent years [19].

Based on that setting, we used Eq. (1) to evaluate the water head required for 1 m^3 of water in a reservoir, which is sufficient for a desalination plant to produce 1 m^3 of desalinated water. The result shows that the required water head reaches 734 m and 1211 m when the specific energy consumptions of desalination are set to be 2.0 and 3.3 kWh m^{-3} , respectively.

$$\begin{aligned} & \text{Power required for } 1 \text{ m}^3 \text{ of desalinated water } (\text{kWh m}^{-3}) \\ & \times 1000 \times 3600 \\ & = 1 \text{ m}^3 \text{ water} \times \text{density of water} \times 9.81 (\text{m s}^{-2}) \\ & \times \text{water head (m)} \end{aligned} \quad (1)$$

To reap the greatest benefit from this model, only reservoirs higher than 734 m were considered; Techí reservoir, as shown in Fig. 5, which is located upstream of Dajia River, was selected. The specifications and the effective storage capacities at the end of 2014 for this reservoir are shown in Table 1.

The designed storage capacity of Techí reservoir represented 9.2% of all reservoirs in Taiwan at the end of 2014. After operating for forty years, about 43% of the capacity of this reservoir is ineffective because of silt deposits brought by typhoons and heavy rains; this problem is common to most Taiwan reservoirs.

As Fig. 6 shows, most reservoir water is used for hydroelectric power only through Techí reservoir and its downstream hydro-power plants in a series, such as units at Chin-Shan, Guguan, Tianlun and Maan; the tail water finally accumulates in the Shigang reservoir to satisfy most water demand. Fig. 7 shows the trends of daily water inflow, outflow, rate of water storage of Techí reservoir [21] during 2011–2014. Fig. 8 shows the average water-demand profile of its service area [22], which is an average value of data from 2011 to 2014. According to the WRA prediction of water supply and demand of Taichung city in 2031 [23], even with the most active policy of water economy, the domestic daily water demand will increase from 1.281 Mm^3 to 1.522 Mm^3 . Under this premise, when considering the water support from the neighborhood reservoir in Miaoli County, the water-demand profile as Fig. 8 shows must still increase by 11%.

As the daily inflow patterns are unavailable for the Chin-Shan, Guguan, Tianlun and Maan reservoirs, the water inflow patterns of these reservoirs are assumed to be the same as for Techí reservoir, shown in Fig. 7, but the annual inflow is corrected with actual

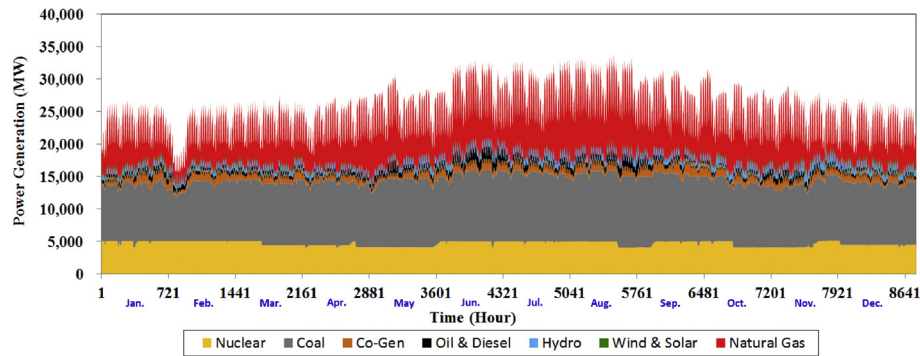


Fig. 3. Hourly power generation in 2011.

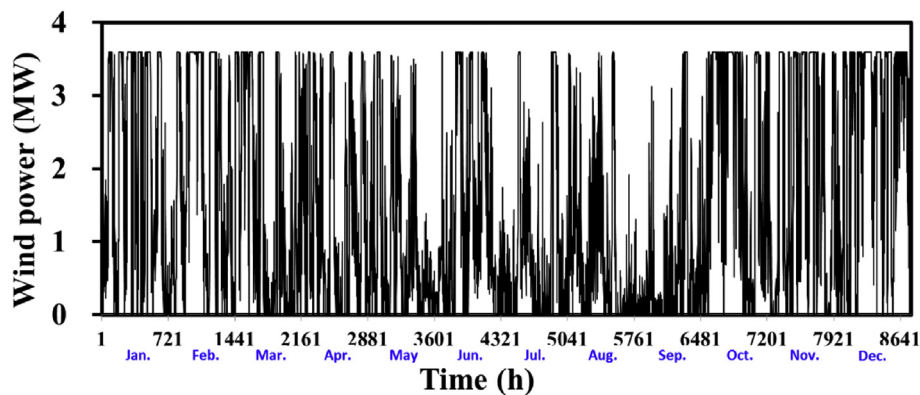


Fig. 4. Simulated power output of a single offshore wind turbine.



Fig. 5. Locations of pilot offshore wind projects and reservoirs selected in this research.

data. As only daily inflow data are available, the distribution of inflow for each hour within a day is assumed to be the daily value divided by 24.

3.4. Case selection

Four inflow patterns shown in Fig. 7 are selected as four cases (cases A, B, C and D) for analysis; each case has its own climatic characteristics. The annual inflow in 2011 (case A) was the least during the past decade; thus case A represents a minute annual inflow case. As we attempt to verify that the proposed mode can survive under a continuous weather pattern, data through 2012 to 2014 are selected as the following cases of case A. The annual inflow in 2012 (case B) is 27% greater than the average value of the past decade; it also represents an enormous rainfall pattern that is related to typhoons. The annual inflow in 2013 (case C) is 95% of the average value of the past decade; this case serves to observe the system response under average climatic conditions. Although the annual inflow in 2014 (case D) is not the least, the minute rainfall during the cold season in 2014 has caused a severe drought in 2015. To test the system response under this scenario is hence meaningful.

For the purpose of comparison, we define a reference case that takes most data from the average value of years 2011–2014.

4. Method

In this research, one year (8760 h in total) was divided into three independent periods denoted T1 ($t < 3600$ h), T2 ($3600 \text{ h} \leq t < 7460$ h) and T3 ($t \geq 7460$ h) under the premises of satisfying the peak-load period in Fig. 3 and leaving appropriate reservoir water in period T3. During periods T1 and T3, the obligation of water supply from Techi reservoir is relieved with the help of desalination. The supplement of water in Shigang reservoir during these periods comes mainly from tributaries downstream of

Table 1
Specifications of selected reservoirs [20].

Reservoir	Full water level/m	Designed storage capacity/Mm ³	Effective storage capacity/Mm ³	Current generation units/MW	Minimum hourly outflow/Mm ³ h ⁻¹
Techi reservoir	1408	262.2	148.8	234 (1140 ^a)	0.0027

^a All generation units downstream of Techi units are included in the analysis; the total capacity originally installed is 1140 MW.

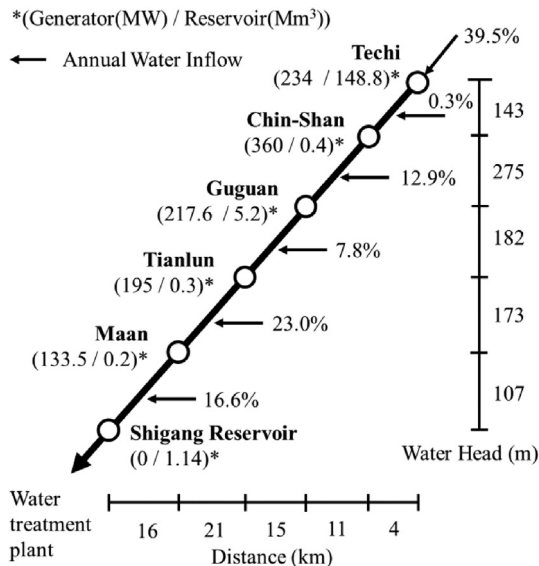


Fig. 6. Hydropower units and annual water inflow distribution of Dajia River (original system).

Techi reservoir. As the plum rain that brings about 30% of annual rainfall in Taiwan generally falls in May and June, the rate of water storage of reservoirs increases rapidly during this period; and time $t = 3600$ is selected as the initiation of hydropower in full operation. The reservoir water will cover most water demand during period T2.

The proposed model is established with programming software (Matlab). The target of the iterative process is to maximize the hydro-turbine capability of peak-hour sharing without causing water shortage under the operating strategies described in Sections 4.1 and 4.2.

4.1. Strategies of system operation during period T1 and T3

Fig. 9 shows how the proposed model selects a specific operating mode for each data input. The data input for each hour includes the inflow of reservoir, water demand, and the wind power. For periods T1 and T3, the system runs the upper half part of Fig. 9 and tests the water balance ($Q_{balance}$) based on Eq. (2):

$$Q_{balance}(t) = Q_{Maan}(t - 6) + Q_{tributary}(t - 3) - Q_d(t) - Q_{eco}(t) \quad (2)$$

in which $Q_{Maan}(t - 6)$ and $Q_{tributary}(t - 3)$ represent the flow released from Maan unit and downstream tributaries of Techi reservoir from six hours and three hours previously, respectively; Q_d and Q_{eco} represent water demand and flow for ecological consideration, respectively. At time t , all hydro-units are operated simultaneously and the same amount of water are released from units of Techi, Chin-Shan, Guguan, Tianlun and Maan, thus $Q_{Maan}(t)$ equals to the water released from Techi reservoir at time t ($Q_{Techi}(t)$). Based on the measured data [24], the time required for Shigang reservoir to

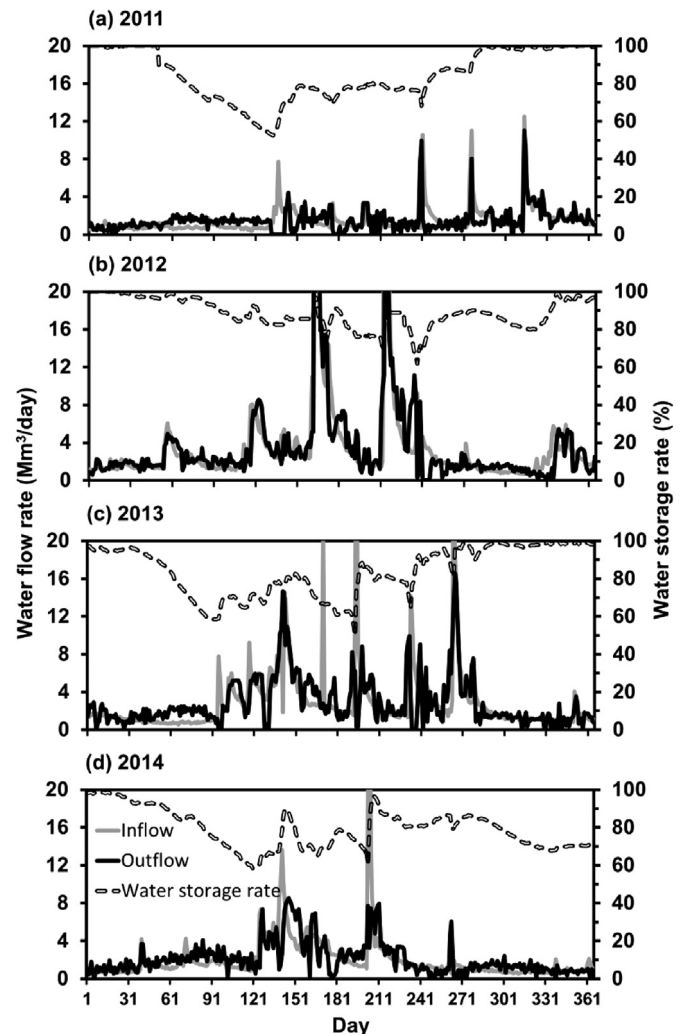


Fig. 7. Inflow, outflow and water storage percentage of Techi reservoir from 2011 to 2014.

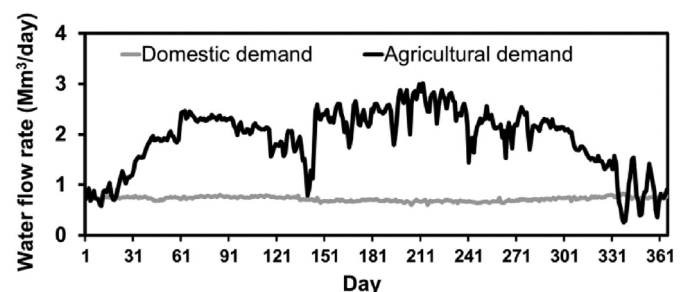


Fig. 8. Profile of average water demand.

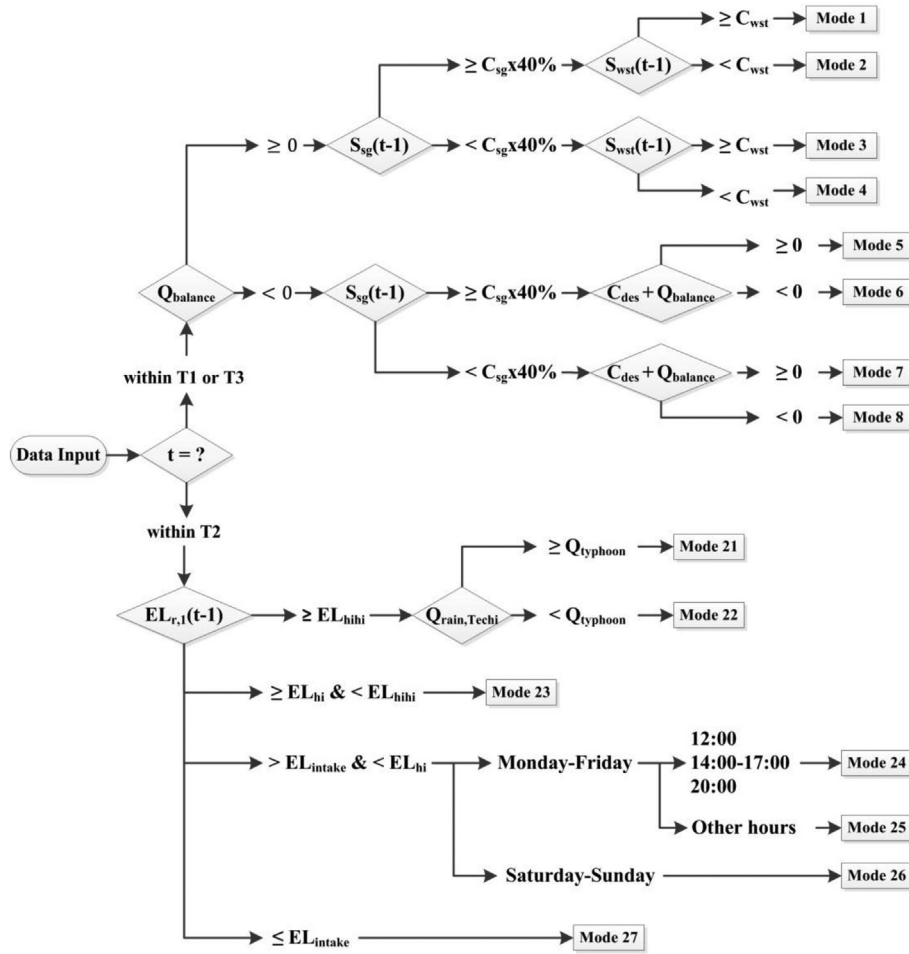


Fig. 9. Logic for mode selection.

receive the water released from Maan unit, which is also defined as the delay time, is about 6 h, and this is why $Q_{Maan}(t-6)$ is used in Eq. (2). Since most of the inflow from downstream tributaries of Tech reservoir comes from the upstream of Maan and Shigang reservoirs (as shown in Fig. 6), we have simplified the release point of incoming flow of tributaries to be the middle point between Shigang reservoir and Maan unit; the delay time under this setting is 3 h. Actually, the sensitivity study of this delay time (ranges from 3 to 12 h) is performed, and the result shows that the effect of delay time is minor due to the buffering effect of Shigang reservoir and it can be neglected.

$Q_{balance}$ determines how desalination and WST are involved for the water supply. If $Q_{balance}$ is positive, it means that the inflow of Shigang reservoir can satisfy Q_d and Q_{eco} . Under this condition, actions for water reservation are possible. The proposed model tests further whether the rate of water storage of Shigang reservoir (S_{sg}) exceeds 40% of its capacity (C_{sg}) and whether the rate of water storage of WST (S_{wst}) is at its full level; four operating modes (modes 1–4) are thus determined for these conditions.

In contrast, if $Q_{balance}$ is negative, the inflow of Shigang reservoir cannot satisfy the aforementioned water demand and action is needed to compensate this shortage. Under this condition, the proposed model tests further whether S_{sg} exceeds 40% of its capacity (C_{sg}) and whether the full operation of desalination plants can cover this shortage ($C_{des} + Q_{balance} \geq 0$); four modes (modes 5–8) are thus determined for these conditions.

Based on various operating modes, the water flow from

desalination plants (Q_{des}) can be zero (Eq. (3)a) for modes 1 and 6, be fully operational (Eq. (3)b) for modes 2, 3, 4, 7 and 8, or be partially operational to satisfy the water shortage (Eq. (3)c) for mode 5. These regulations are designed to preserve the reservoir water as much as practicable, but some special conditions alter the applicable equations for some modes. For example, if flooding is possible for charging WST, mode 2 turns to use Eq. (3)e to produce just enough desalination water to equal the residual space in WST; if flooding is possible in Shigang reservoir due to upstream inflow, modes 3 and 4 stop the desalination plants (Eq. (3)a); if the residual space of Shigang reservoir is sufficient for excess inflow of $Q_{balance}$ but is insufficient when the desalination plants share part of the water demand in a fully operational mode, modes 3 and 4 turn to use Eq. (3)d to produce just enough desalination water to keep the Shigang reservoir at its full level.

To charge WST, if part of the desalination flow can be used to charge WST without flooding in WST, mode 5 uses Eq. (3)b to satisfy the shortage and to charge WST at the same time, but, if flooding in WST is possible, mode 4 turns to use Eq. (3)d plus Eq. (3)e and mode 5 turns to use Eq. (3)c plus Eq. (3)e to satisfy the shortage and to charge WST at the same time.

$$Q_{des}(t) = \begin{cases} 0 & a \\ C_{des} & b \\ -Q_{balance}(t) & c \\ C_{sg} - S_{sg}(t-1) - Q_{balance}(t) & d \\ C_{wst} - S_{wst}(t-1) & e \end{cases} \quad (3)$$

The discharged flow of WST ($Q_{wst,d}$) is defined by Eq. (4)a–c. For modes 1–5 and 7, the operation of WST is not required (Eq. (4)a). For modes 6 and 8, depending on the remaining water in WST, the maximal supply (Eq. (4)b) or (Eq. (4)c) of water is selected.

$$Q_{wst,d}(t) = \begin{cases} 0 & a \\ C_{des} & b \\ S_{wst}(t-1) & c \end{cases} \quad (4)$$

The flow charged to WST ($Q_{wst,c}$) is defined by Eq. (5)a–d. Generally, only mode 2 charges WST with all water produced by desalination plants (Eq. (5)b); other modes do not charge WST (Eq. (5)a), but, when the water demand is satisfied and excess water is produced by the desalination plant, modes 4 and 5 turn to use Eq. (5)c if the residual space of WST is small or to use Eq. (5)d if the residual space of WST is large.

$$Q_{wst,c}(t) = \begin{cases} 0 & a \\ Q_{des}(t) \times 1 \text{ h} & b \\ C_{wst} - S_{wst}(t-1) & c \\ C_{des} - Q_{des}(t-1) \times 1 \text{ h} & d \end{cases} \quad (5)$$

The remaining water in WST (S_{wst}) and Shigang reservoir (S_{sg}) are then determined by the balance calculation of flow in Eqs. (6) and (7), respectively.

$$S_{wst}(t) = S_{wst}(t-1) + (Q_{wst,c}(t) - Q_{wst,d}(t)) \times 1 \text{ h} \quad (6)$$

$$S_{sg}(t) = S_{sg}(t-1) + (Q_{balance}(t) + Q_{des}(t) + Q_{wst,d}(t)) \times 1 \text{ h} \quad (7)$$

After treating the interactions among Shigang reservoir, desalination, WST and water demand, we consider those parts related to Tech reservoir and hydroelectric power. The water through the hydro-turbine (Q_{hydro}) of Tech reservoir is determined by Eq. (8) a–b. The hydro-turbine is generally not operated (Eq. (8)a) for modes 1–6 to preserve as much reservoir water as practicable, but is fully operated (Eq. (8)b) for modes 7–8 to compensate the low water level in Shigang reservoir. If the rate of water storage of Tech reservoir (S_{Tech}) exceeds 98% capacity, modes 1–6 turn to use Eq. (8)b for the safety concern.

$$Q_{hydro}(t) = \begin{cases} 0 & a \\ Q_{hydro,max}(EL_{r,1}(t), Q_{hydro,design}, X_{pattern}) & b \end{cases} \quad (8)$$

The calculation of maximal flow through hydro-turbine ($Q_{hydro,max}$) in Eq. (8)b is an iterative process and is related to the water level of Tech reservoir ($EL_{r,1}$), the designed flow of hydro-turbine ($Q_{hydro,design}$) and output ratio of hydropower in a specific hour ($X_{pattern}$). The value of $X_{pattern}$ ranges from 0 to 1, depending on the period, as shown in Table 2, to prevent an unnecessarily large Q_{hydro} during a non-peak-load period, but $X_{pattern}$ equals 1 if S_{Tech} exceeds 98%.

Table 2
Values of $X_{pattern}$.

Period	Hours	Mon. – Fri.	Sat.	Sun.
T1 & T3	All	0–0.13	0–0.13	0–0.13
T2	11:00 a.m.–12:00 a.m.	0.3	0	0
	1:00 p.m.–2:00 p.m.	1	0.6	0.9
	2:00 p.m.–3:00 p.m.	0.85	0	0
	3:00 p.m.–4:00 p.m.	0.65	0	0
	4:00 p.m.–5:00 p.m.	0.2	0	0
	7:00 p.m.–8:00 p.m.	0.5	0.5	0.6
	11:00 p.m.–12:00 p.m.	0	0.75	0.97
	the others	0	0	0

The aforementioned iterative process begins from an initial guess of $Q_{hydro,max}$ that is restricted by the product of $Q_{hydro,design}$ and $X_{pattern}$; the outflow of Tech reservoir (Q_{Tech}) is determined with Eq. (9):

$$Q_{Tech}(t) = Q_{hydro}(t) + Q_{eco} + Q_{flushing}(t) + Q_{desilting}(t) \quad (9)$$

in which Q_{eco} is the constant outflow for ecological consideration; $Q_{flushing}$ and $Q_{desilting}$ are the flow rates for flushing and desilting for safety concerns, respectively, but are not used during periods T1 and T3.

With known values of Q_{Tech} from Eq. (9) and the inflow of Tech reservoir ($Q_{rain,Tech}$) from input data, the water amount (S_{Tech}) and water level ($EL_{r,1}$) of Tech reservoir are determined with Eqs. (10) and (11), respectively.

$$S_{Tech}(t) = S_{Tech}(t-1) + Q_{rain,Tech}(t) \times 1 \text{ h} - Q_{Tech}(t) \times 1 \text{ h} \quad (10)$$

$$EL_{r,1}(t) = -4 \times 10^{-7} \times S_{Tech}(t)^3 - 0.001 \times S_{Tech}(t)^2 + 0.5321 \times S_{Tech}(t) + 1352.3 \quad (11)$$

The values of Q_{hydro} and $EL_{r,1}$ are then substituted into Eq. (12) to calculate the hydropower output of Tech reservoir ($E_{r,1}$):

$$E_{r,i}(t) = \phi_{hydro,i}(t) \times Q_{hydro}(t) \times [EL_{r,i}(t) - EL_{disc,i}(t)] \times 9.81 \times [1 - \eta_{loss}] \times 1000/3600 \quad (12)$$

in which i represents the hydro-turbine units in calculation, and 1 through 5 denote units for Tech, Chin-Shan, Guguan, Tianlun and Maan, respectively; $\phi_{hydro,i}$ represents the efficiency of each unit; $EL_{r,i} - EL_{disc,i}$ represents the water head of each unit as shown in Fig. 6; η_{loss} represents the head loss and is assumed to be 5%.

Rather than assuming a constant efficiency of a hydro-turbine, a variable efficiency with a decreased flow rate [25] is used during the power calculation of Tech units when the water level is above the design point. The output of hydropower obtained from Eq. (12) is then compared with the installed capacity of Tech's hydropower ($C_{hydro,1}$). If $E_{r,1}$ is greater than $C_{hydro,1}$, a smaller value of Q_{hydro} is estimated and iterated again until $E_{r,1}$ is smaller than $C_{hydro,1}$.

The power consumed by desalination plants (E_{des}) is determined with Eq. (13):

$$E_{des}(t) = Q_{des}(t) \times \phi_{des} \times 1000 \quad (13)$$

in which ϕ_{des} represents the specific energy consumption of desalination and is set as 3.3 kW h m^{-3} .

Finally, the proposed model calculates the required power support from CGT (E_{cgt}) as shown in Eq. (14)a–b. If the wind power generated at time t (E_{wind}) is greater than E_{des} or no desalination plant is operated, no power support from CGT is needed (Eq. (14)a); otherwise, power support is required in Eq. (14)b.

$$E_{cgt}(t) = \begin{cases} 0 & a \\ E_{des}(t) - E_{wind}(t) & b \\ E_{des}(t) & c \end{cases} \quad (14)$$

4.2. Strategies of system operation during period T2

The lower half of Fig. 9 shows how to identify modes 21–27

during period T2. As there is typically abundant rainfall during T2, the safety concern about the water level in Tech reservoir ($EL_{r,1}$) becomes important. To protect the reservoir and to support the peak-hour demand, three water levels EL_{hihi} , EL_{hi} and EL_{intake} are designated.

When $EL_{r,1} \geq EL_{hihi}$, the water level is approaching full; the system tests whether the inflow of Tech reservoir ($Q_{rain, Tech}$) is greater than the criterion of flushing action ($Q_{typhoon}$) to determine modes 21 and 22. When $EL_{hi} \leq EL_{r,1} \leq EL_{hihi}$, a flushing action is not required for any inflow conditions, but the hydro-turbine units must be fully operated to help to lower the water level. This condition is defined as mode 23.

$EL_{r,1}$ is typically between the elevation of the hydropower intake duct (EL_{intake}) and EL_{hi} , and three modes 24–26 are designed for peak hours during Monday and Friday, non-peak hours during Monday and Friday, and the period Saturday and Sunday, respectively.

When $EL_{r,1} \leq EL_{intake}$, the water level is low and hydropower is not available; mode 27 is designed for this condition to support the water demand only.

The calculations of modes 21–27 are similar to those in Section 4.1, and only those parts with other considerations are described below.

The desalination plants are generally not operated (Eq. (3)a) for modes 21–24 and 27, and are fully operated (Eq. (3)b) for mode 26. Even so, mode 26 stops the desalination plants (Eq. (3)a) if S_{sg} is above its 90% of C_{sg} and $Q_{tributary}$ is larger than the water demand. Mode 25 is particularly designed to compensate a possible water shortage or to recover the water level of WST; desalination plants are fully operated to support the water demand when S_{sg} is below 10% of C_{sg} or it turns to charge WST with Eq. (3)a, b and e (depending on the residual space of WST) when S_{sg} is above 10% of C_{sg} .

The WST is generally not operated (Eqs. (4)a and (5)a) for all modes, but if S_{sg} has a negative value without WST support, WST is either fully (Eq. (4)b) or partially discharged (Eq. (4)c) for modes 24 and 26 that depends on the remaining water in WST. When S_{sg} is above 10% of C_{sg} , mode 25 turns to charge WST with Eq. (5)b.

The hydro-turbine is not operated (Eq. (8)a) for modes 21, 25 and 27 because of a maximal flushing requirement, non-peak-hour, and minute water level, respectively, but is operated (Eq. (8)b) for modes 22–24 and 26 to satisfy the power demand. For safety concerns, when applying Eq. (9), flushing and desilting actions are required for mode 21, and flushing action is required for mode 22. $X_{pattern}$ follows the regulation stated in Table 2 for modes 24 and 26, and equals 1 for other modes having a hydropower output.

The calculations of other parameters not mentioned here resemble those in Section 4.1.

4.3. Parameter settings and assumptions

The key parameters involved in the calculations are shown in Table 3. Most parameters are based on the actual designed values or operating guidelines of existing reservoirs and hydro-turbine units; the parameters related to the desalination plants are based on the announced desalination project as described in Section 3.3, but some assumptions are made for the construction of the proposed model, described as follows.

By the end of 2014, the effective capacity of Shigang reservoir was only 62% of the designed capacity. As WRA continues to recover effective capacity of Shigang reservoir in recent years – for example, 0.3 Mm³ was recovered in 2015 – we reasonably assume that the effective capacity can be recovered to 92% (1.7 Mm³) of its designed capacity by 2030. With this assumption, the required capacity of WST is greatly decreased. The initial levels of water in all reservoirs and WST are assumed to be their full levels.

Table 3
Descriptions and settings of system parameters.

Parameter	Description	Value
Shigang reservoir		
C_{sg}	Shigang reservoir effective capacity/Mm ³	1.7
Tech reservoir		
C_{Tech}	Tech reservoir effective capacity/Mm ³	148.8
EL_{full}	Reservoir full water level/m	1408
$EL_{noflush}$	Below this water level, flushing is not possible/m	1347
EL_{hi}	Above this elevation, hydropower is fully operated/m	1406
EL_{hihi}	Above this elevation, flushing is required/m	1407
EL_{intake}	Elevation of Tech hydropower intake duct/m	1350
$EL_{disc,1}$	Elevation of Tech hydropower discharge duct/m	1241
Q_{eco}	Outflow for ecological consideration/Mm ³ h ⁻¹	0.0054
$Q_{typhoon}$	Inflow above this limit requires flushing activity/Mm ³ h ⁻¹	2.52
$Q_{desilting}$	Desilting flow rate/Mm ³ h ⁻¹	0.54
$Q_{flushing}$	Flushing flow rate/Mm ³ h ⁻¹	1.08
Hydro-turbine units		
$C_{hydro,1}$	Proposed Tech hydropower capacity/MW	269
$C_{hydro,2}$	Proposed Chin-Shan hydro-turbine capacity/MW	457
$C_{hydro,3}$	Proposed Guguan hydro-turbine capacity/MW	310
$C_{hydro,4}$	Proposed Tianlun hydro-turbine capacity/MW	269
$C_{hydro,5}$	Proposed Maan hydro-turbine capacity/MW	176
$Q_{hydro,design}$	Proposed Tech hydropower flow at design condition/Mm ³ h ⁻¹	0.6858
H_{design}	Proposed Tech hydropower designed water head/m	160
η_{loss}	Proposed hydro-turbine head loss/%	5
$\phi_{hydro,1}$	Proposed Tech hydropower efficiency at design point/%	90
$\phi_{hydro,2}$	Proposed Chin-Shan hydro-turbine efficiency at design point/%	91
$\phi_{hydro,3}$	Proposed Guguan hydro-turbine efficiency at design point/%	91
$\phi_{hydro,4}$	Proposed Tianlun hydro-turbine efficiency at design point/%	87
$\phi_{hydro,5}$	Proposed Maan hydro-turbine efficiency at design point/%	88
Desalination plants		
C_{des}	Desalination plants' capacity/Mm ³ h ⁻¹	0.013
C_{wst}	WST capacity/Mm ³	0.31
ϕ_{des}	Specific energy consumption of desalination/kWh m ⁻³	3.3

The values of EL_{hi} and EL_{hihi} for the operation of Techii reservoir is determined through an iterative process to fulfill the demands of water and peak-hour power. An increased capacity 342 MW of a Francis-type hydro-turbine, which is the maximal capacity added without causing a water shortage during calculations, is proposed in this research to enhance the capability of power supply during peak hours. As the existing Techii hydropower units have operated over 40 years, both the proposed model and reference case are assumed to be retrofitted with new hydropower units of increased efficiency (original $\phi_{hydro,1}$ is 76.6%) and designed water head (H_{design} is increased by 12%), able to generate the same power with less water ($Q_{hydro, design}$ is decreased by 12.4%). As the hydropower units downstream of Techii reservoir receive and release the same amount of water within one hour of operation, the efficiencies of the hydro-turbines are assumed to be the designed values for simplicity.

5. Results and discussion

5.1. Water supply

When compared with the actual reservoir outflow patterns in Fig. 7, the proposed model shows a more regular pattern for the reservoir operations in the left part of Fig. 10. After the operation of period T2, the rates of water storage decrease greatly in cases A and D, which are the small annual inflow cases as shown in Table 4. An

increased hydropower output at all peak hours is impracticable without the support of the proposed desalination plants as shown in the right part of Fig. 10.

The rate of water storage is an index to test the system sustainability. The system is identified as a sustainable system if the final water storage rate is about the same as the initial value. Cases A–C all satisfy this criterion; case D apparently fails this criterion, but this result does not mean that the proposed model is unfeasible. By the end of 2014, Taiwan had suffered the most severe drought since 2002; the rate of Techii water storage decreased to 73.1% as shown in Fig. 7. Although the proposed model would further decrease the rate of water storage to 50.7%, the desalination plants can compensate this 22.4% (34.3 Mm³) deviation within 109 days, and maintain a minimum water supply even when the reservoir is empty. The rising curve of rate of water storage for the end of the year according to case D is also evidence that an ensured source of water (desalination) in the proposed model is beneficial for a drought year.

The seventh row of Table 4 shows that the fraction of water utilized for hydropower decreases greatly when flushing is required for safety concern. In this study, no prediction of a typhoon is included; a possible benefit from a large inflow brought with a typhoon is hence neglected in the proposed model. The proposed model ensures a minimum flow 1.5 m³ s^{−1} for ecological consideration at any time, which is larger than the actual operating value 0.75–1.41 m³ s^{−1} from Techii reservoir during periods T1 and T3

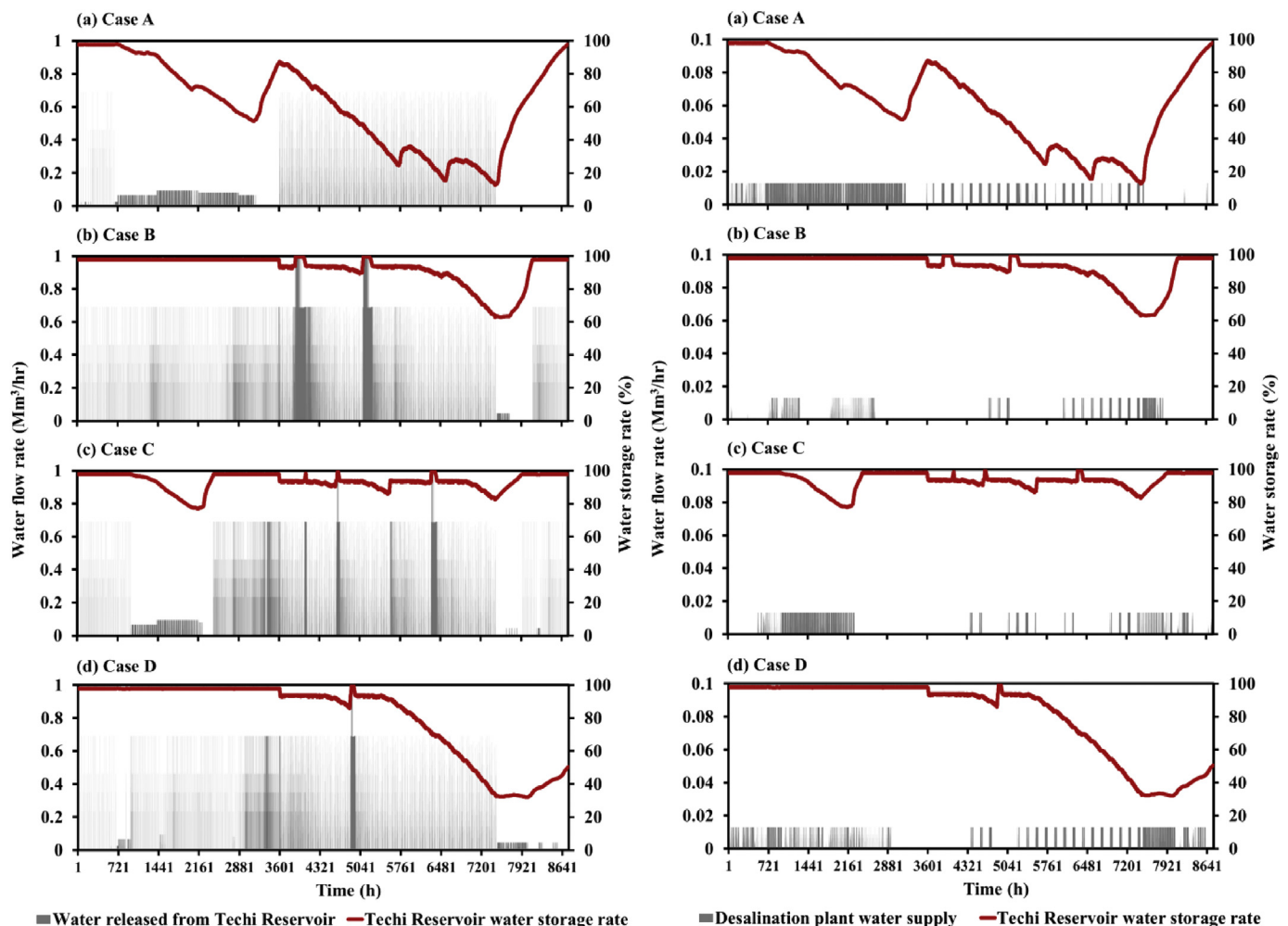


Fig. 10. Patterns of water supply of Techii reservoir (left) and desalination plant (right).

Table 4
Annual water supply for each case.

Descriptions	Case A	Case B	Case C	Case D	Average	Reference
Water sector						
Techi reservoir inflow/Mm ³	547	1557	1155	835	1024	1024
Water demand/Mm ³	993	993	993	993	993	993
Techi reservoir outflow/Mm ³	548	1557	1155	906	1041	1031
Desalination water/water demand	4.4%	1.5%	2.5%	3.2%	2.9%	0.0%
Water for hydropower/reservoir outflow	91.4%	77.5%	93.3%	92.9%	87.0%	93.6%
Water for ecological flow/reservoir outflow	8.6%	3.0%	4.1%	5.2%	4.5%	–

[24], as shown in the last row in Table 4.

5.2. Power supply

Fig. 11 compares the patterns of hydropower generated according to the proposed model with the assumed pattern of wind power. During period T2, all cases deliver maximal hydropower at peak hours, such as 11–12 a.m., 1–5 p.m., and 7–8 p.m.; the period of peak power demand during summer is well covered. In contrast, the hydropower supplies show different patterns among all cases during periods T1 and T3, which depend mostly on the inflow amount and distribution in these periods. For example, the inflow during period T1 is small in case A, and the water in Tech reservoir is preserved for subsequent operation during T2; for this reason little hydropower is produced during period T1. The small inflow

during period T1 would cause the rate of reservoir water storage to be too small after T2 operation, and would further limit the hydropower operation during period T3.

According to the results shown in Table 5, the average power consumption of desalination plants is only 96 GWh, of which 69.8 GWh occurs during periods T1 and T3. To understand how the proposed system consumes wind power during periods T1 and T3, the relations among consumption of wind power, consumption rate of wind power (wind power consumed over wind power produced), and installed capacity of wind power are investigated in Fig. 12. The power consumption of wind power increases greatly with the increased number of wind turbines, but becomes saturated when 540 MW of wind turbines are installed. On the contrary, the consumption rate of wind power decreases with an increased number of wind turbines, and 42%, 26% and 19% of wind power produced are consumed by the desalination plants when 36 MW, 72 MW and 108 MW of wind turbines are installed, respectively. This result provides that, when the installed capacity of desalination plants and wind turbines are comparable, about 40% of wind power produced during periods T1 and T3 is consumed to enhance the hydropower output during T2.

One purpose of the proposed model is to enhance the ability of peak-load support for the existing hydropower units; the power generated according to the proposed model is compared with the entire power demand in Taiwan in 2011, shown in Fig. 13. With 342 MW of new hydro-units added, cases A, C and D effectively share the peak load 1368 MW h, but case B shows a capability only 803 MW h because a typhoon was identified and hydropower was unavailable when much flushing was required during hours between 5113 and 5160. As the power demand profile is taken from 2011, this profile might not relate to the actual events of a specific year. As the peak power demand does not occur during a typhoon period in reality, we reasonably omit the questionable behavior of the proposed model during these periods. The hydropower of the proposed model is hence still able to take the peak power demand, 1368 MW h. For the purpose of comparison, it is interesting to understand the behavior of the original hydropower system (reference case) under a peak-load period. In Fig. 14, the original hydropower system takes a peak power demand only 389 MW h despite its large installed capacity; this fact is not astonishing as the operation of hydropower depends mostly on the water supply duty of the reservoir. The data of hydropower generation in Fig. 14 are for the entire traditional hydropower system of TPC; the actual capability of covering a peak load is hence even less than 389 MW h for the reference system.

Regarding the total hydropower produced, the hydropower for the proposed model ranges between 991 GW h and 2488 GW h with average value 1852 GW h; the reference case is 1981 GW h. The proposed model decreases the hydropower generation by 6.5% for several reasons, such as greater reservoir outflow that does not pass hydropower units for ecological considerations and no pre-release of reservoir water to accommodate increased inflow from a typhoon in the proposed model. These trends are also observed in the seventh row in Table 4, in which the fraction of water for

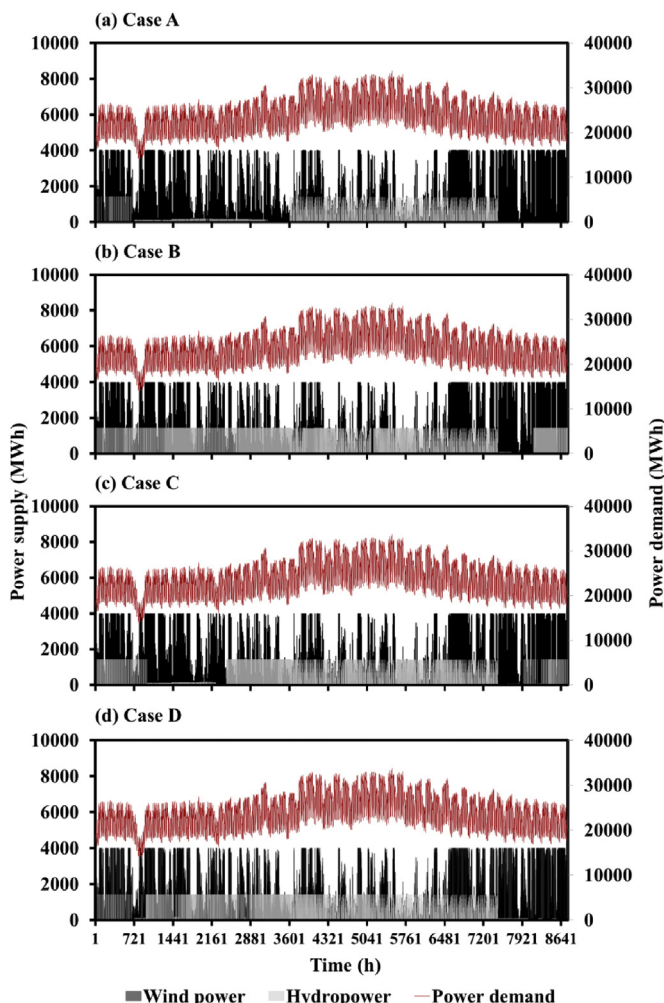


Fig. 11. Patterns of power supply of the proposed model.

Table 5
Annual power production for each case.

Descriptions	Case A	Case B	Case C	Case D	Average	Reference
Power sector						
Power consumed by desalination plants/GWh	145	50	83	106	96	147
Power produced by hydropower/GWh	991	2488	2204	1724	1852	1981
Total generating capacity/MW	1482	1482	1482	1482	1482	1140
Added hydropower capacity/MW	342	342	342	342	342	—
Added CGT capacity/MW	—	—	—	—	—	979
Peak-load sharing/MW	1368	803	1414	1417	1251	1368 ^a

^a After adding 979 MW of new CGT units, the capability of peak power sharing increases from 389 MW to 1368 MW.

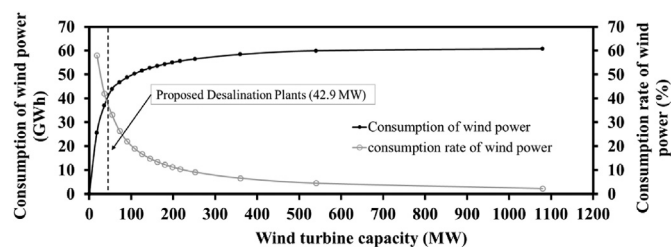


Fig. 12. Variation of wind power consumption and rate of consumption with wind turbine capacity during periods T1 and T3.

hydropower over reservoir outflow is 6.6% less in the proposed model. As these factors make the evaluation of the proposed model conservative, we retained these assumptions when a benefit analysis was performed.

5.3. Benefit analysis

Before an analysis of the total benefits of the system, the boundaries of both systems must be expanded to support an additional annual water supply the same as that in the new reservoir of the reference case, the minimum peak-load sharing capability as that in the proposed model and the same annual power generated as that in the reference case. In Table 6, these assumptions lead to total water production 95 Mm³ in desalination plants, even though the proposed model requires only 29 Mm³. The addition of 342 MW and 979 MW generating units for the proposed model and the reference case, respectively, and an additional power supply 436 MW h with the existing combined cycle (CC) power units are in the proposed model. The capital cost and operating and maintenance (O&M) costs for a desalination plant and a new reservoir come from a WRA feasibility study of desalination plants in the Taiwan coastal area [26,27] and a feasibility study of Tien Hwa Hu reservoir [28–30] with some modifications of economic parameters and the additional cost of water purification in this work. Most cost data for additional CGT plants and hydropower units are taken from a study by USA Energy Information Administration (EIA) [31,32], but the capital cost of hydropower units in the proposed model is taken from the rehabilitation project of the Chin-Shan hydropower plant, which was totally destroyed by a typhoon and rehabilitated by the end of 2015 [33]. Under discount rate 5.0%, inflation rate 2.0% and 1.6% for O&M and LNG price, the levelized cost of water production in desalination plants and a new reservoir are NT\$30.6 m⁻³ and NT\$20.6 m⁻³, respectively. This effect would cause the proposed model to expend NT\$955 million more than the reference case for the additional water production per year, but the leveled cost of power production with added units to meet the peak-load requirements are NT\$9.21 kW⁻¹ h⁻¹ and NT\$12.27 kW⁻¹ h⁻¹ for the proposed model and reference case, respectively, which would cause the reference case to expend

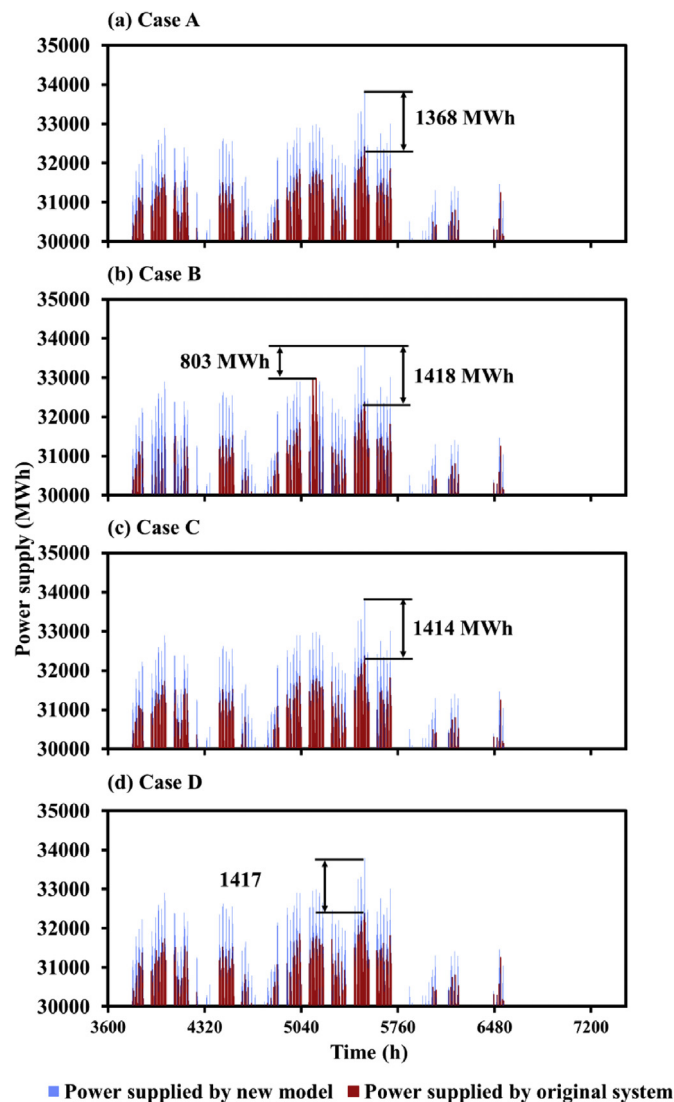


Fig. 13. Ability of peak power sharing of the proposed model.

NT\$1364 million more than the proposed model for the additional peak-load support per year. In summary, the costs for additional water and power are NT\$5318 million and NT\$5727 million per year for the proposed model and the reference case, respectively; the proposed model has a cost saving NT\$409 million per year. With the net positive profit generated by the proposed mode, a large-scale desalination plant, which is generally cancelled after an economic evaluation, becomes practical in Taiwan. During a drought year, the reservoirs in Taiwan typically confront concurrently a water shortage; the new reservoir in the reference case hence does

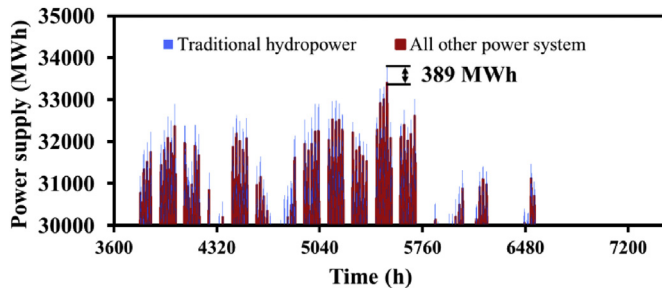


Fig. 14. Ability of peak power sharing of the reference case.

not ensure water safety under this condition. Besides, the existing Techí reservoir is suffering serious silt deposits; the requirement of continuous operation for the entire year makes it difficult to handle this problem. With the support of the desalination plants in the proposed model, these problems can be greatly relieved.

6. Conclusion

We have proposed a new operating model involving reservoir, hydropower, desalination plant and wind power to solve issues of power and water supply, and we demonstrated a case study of Taichung city in Taiwan. The results of our analysis show that the proposed model fulfills the water requirements of Taichung in 2030; the peak-load sharing capability of the existing hydropower units has been greatly increased with only a few additional units. The desalination plants increase the power consumption during the cold season, during which wind power is abundant but the power demand is small, and release the water supply burden of the

existing reservoir to enable full operation during summer peak hours. From the results of the analysis, we delineate several conclusions and benefits of the proposed model as follows.

- With the proposed model, new hydropower units only 342 MW are required to ensure a minimum power supply 1368 MW h during peak hours, whereas the reference case would require an additional 979 MW CGT units to achieve the same capability.
- The peak load typically occurs during hot weather, which decreases the efficiency and output of thermal power plants. The proposed model uses hydropower units only during peak hours; the hot ambient conditions will not affect the power output and efficiency. When considering this benefit, the security of power supply is enhanced with the proposed model.
- Even though the cost of water production from desalination plants is still large in the proposed model, the greater expense NT\$955 million per year from the water sector is compensated with a smaller expense NT\$1364 million per year from the power sector; the proposed model saves NT\$409 million per year in total.
- As the water price in Taiwan is only NT\$11 m⁻³ [34], which is much less than elsewhere in the world, it is impractical to promote large-scale desalination plants to solve a future shortage of water when considering only the price of water. The proposed model takes into account this aspect and solves it on extracting the benefit from the power sector; a new reservoir in the reference case, which is under serious protest in reality, is no longer required.

From the lessons learned in this work, desalination plants can be an effective medium to connect the water and power sectors and to solve their individual issues. As the cost of desalination is

Table 6
Parameters set for benefit analysis.

	Parameters	Proposed Model	Reference
Financial setting	Discount rate	5.0%	5.0%
	Inflation rate of O&M	2.0%	2.0%
	Inflation rate of LNG fuel price	1.6%	1.6%
Water sector	Type of additional water supply unit	desalination plants	reservoir
	Capacity of additional water supply units/Mm ³ h ⁻¹	0.0130	0.0109
	Annual water production by added units/Mm ³	95	95
	Additional water supplied to target system/Mm ³	29	37
	Capacity factor	83.8%	100.0%
	Capital cost Million NT\$/Mm ³ h ⁻¹	1,288,889	2,258,631
	Specific energy consumption for desalination/kW h m ⁻³	3.3	0.0
	O&M cost/NT\$ m ⁻³	6	5
	Cost recovery period/an	20	50
	Type of Additional generating units/MW	hydropower	CGT
Power sector	Capacity of additional generating units/MW	342	979
	Total hydropower capacity/MW	1482	1140
	Total system capacity/MW	1482	2119
	Peak power production by additional units during T2/GW h	135	307
	Annual power production by the hydropower units/GW h	1852	1981
	Annual power requirement from CC plants during non-peak period/GW h	436	0
	Total power production of the system/GW h	2287	2287
	Capacity factor of added units	4.5%	3.6%
	Capital cost of added units/million NT\$ MW ⁻¹	47.1	30.2
	Fixed O&M cost of added units/million NT\$ MW ⁻¹ an ⁻¹	0.44	0.23
Cost analysis	Variable O&M cost of added units/million NT\$ MW ⁻¹ h ⁻¹	0	0.00048
	Fuel cost/NT\$ kW ⁻¹ h ⁻¹	—	3.66 ^a
	Cost recovery period/an	30	30
	Levelized cost of water produced by new system/NT\$ m ⁻³	30.6	20.6
	Additional cost for water per year/million NTD an ⁻¹	2916	1961
	Levelized cost of electricity produced by added units/NT\$ kW ⁻¹ h ⁻¹	9.21	12.27
	Generating cost of existing natural gas fired units/NT\$ kW ⁻¹ h ⁻¹	2.67	2.67
	Additional cost for electricity per year/million NT\$ an ⁻¹	2403	3767
	Additional cost for water and electricity per year/million NT\$ an ⁻¹	5318	5727

^a As no fuel cost data are available for CGT only, this value is estimated from the historical data of all units fired with natural gas.

decreasing, for countries having a coastline, hydropower and wind power but facing issues of water shortage, the time is ripe to re-evaluate the possible benefit brought by desalination plants as demonstrated by this research.

Absolutely, there is still future work to enhance the benefits of the proposed model. For example, the desalination plants can be largely scaled to serve the areas nearby Taichung city, to consume more wind power in the cold season, and to allow greater hydro-power support during peak hours; pumped-hydro turbines can be incorporated into the proposed model to evaluate a more flexible and mutually beneficial system; an optimization of system operation is required to harness as much rainfall from typhoons as practicable to increase profits.

Acknowledgment

Ministry of Science and Technology of Taiwan (ROC) partially supported this work under contract MOST 105-3113-F-002-003.

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